



Envision Energy
Unifers Power Plant Controller
PSCAD Model User Manual

Version 1.1.5b

2026-7-31

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1 Introduction

The Envision Energy EMS Power Plant Controller (PPC) PSCAD model is a simulation model developed in PSCAD/EMTDC simulation software, used with Envision Energy wind turbine or BESS PCS PSCAD model to support grid-connection simulation studies.

This document focuses on the model usage and the control functions with adjustable parameters of the PPC model in PSCAD.

2 PPC Model Overview

The PPC model is provided within one PSCAD file together with Envision Energy WTG or PCS model as shown in Figure 1 below, user could set parameters accordingly in the model menu by double clicking on the module. The model is compatible with PSCAD V5.0.0 and later versions and supports the Intel Fortran Compiler XE 15 and above. Model package folder name and model file name indicates if it is compatible with 32bit or 64bit compiler.

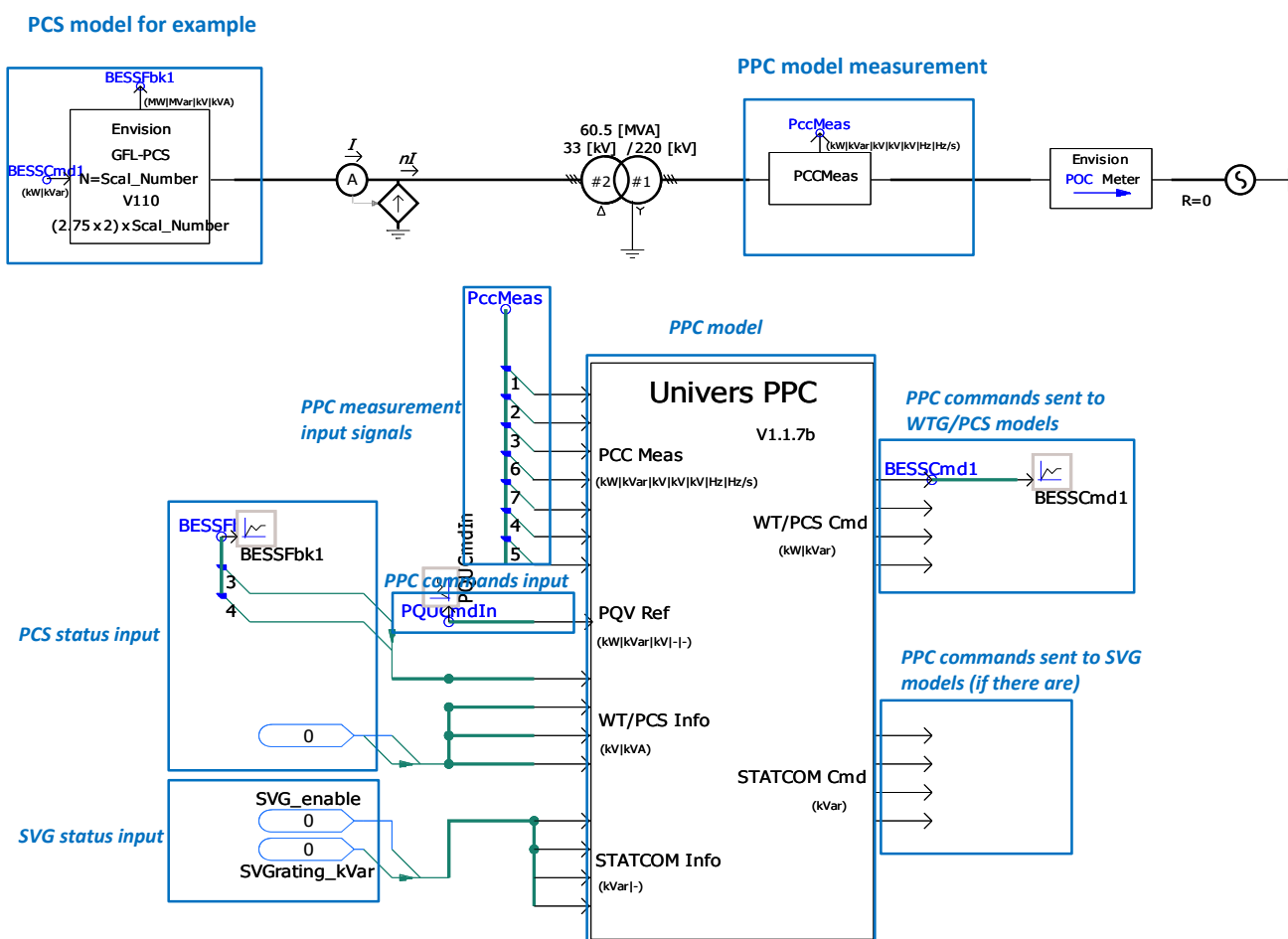


Figure 1 PPC model representation in PSCAD

The PPC model is able to control the active power and reactive power or voltage of its control point and regulate the power to respond to frequency change with “PPC Enable” switched to Enable, or either to work as an EMS dispatcher, bypassing all the closed controllers and directly dispatch active power and reactive power commands to the WTG or PCS models, with “PPC Enable” switched to Disable.

Up to four lumps of WTG or PCS can be controlled by the PPC model. In case more interface is required for plant simulation, please contact Envision Energy to discuss.

The PPC model also provides SVG control function and relevant interfaces; however, please confirm with Envision Energy on product feasibility for SVG control before conducting simulation studies.

3 PPC Model Inputs and Outputs

3.1 Model Inputs

3.1.1 Measurement Inputs

A PPC measurement module, as shown in Figure 2 below, has been prepared by Envision to provide necessary measurement signals to the PPC model.

The module will provide real value measurements to the PPC model to conduct control functionalities. User needs to locate the measurement module to the PPC control point and doesn't need to any setting; please keep the “PCC_Measure” signal name unchanged so that it can be read by PPC.

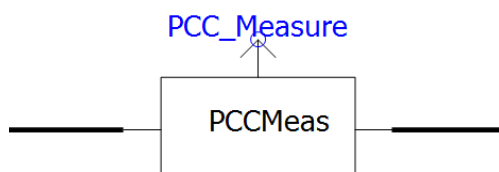


Figure 2 PPC measurement module

3.1.2 Command Inputs

The PPC model reads commands via the multi-dimension signal “PQUCmdIn”. It reads PPC signals in sequence as below:

- 1) Active power setpoint in kW,
- 2) Reactive power setpoint in kVA,
- 3) Voltage setpoint in kV,
- 4) Power factor setpoint.

3.1.3 Controlled Equipment Information Inputs

To have the PPC model correctly calculate controlled capacity and have status of controlled lumps, PPC needs to read terminal voltage and total capacity of each aggregated WTG or PCS model. This is

achieved via reading multi-dimension signal “WTfbk” from WTG model output or “BESSfbk” from PCS model output. If more WTG or PCS model lumps are to be controlled by the PPC model, user needs to set the input signal for the additional WTG or PCS model accordingly. Each of the WTG/PCS inputs is a two-dimension signal including in sequence terminal voltage in kV and lump capacity in kW.

3.2 Model Outputs

To each WTG or PCS model, the PPC model outputs a two-dimension signal “WTCmd” or “BESSCmd”, including active power and reactive power commands to the aggregated WTG or PCS lump, in kW and KVA. If more WTG or PCS model lumps are to be controlled by the PPC model, user needs to set the output signal for the additional WTG or PCS model accordingly.

4 PPC Model Control Functions and Parameters

4.1 General Configurations

In sub-menu “General”, user could configure the basic and general settings of the PPC model. Table below describes the parameters.

Table 1 PPC general settings description

Parameter	Description	Remark
PPC enable	Enable = regular master PPC Disable = EMS dispatcher*	User to configure accordingly
PPC Initialization Time [s]**	Time to start PPC regulation, considering that PPC regulation should not start until WTG or BESS model is initialized	Not adjustable
WT/BESS Terminal Rated Voltage [kV]	Controlled WT or BESS PCS rated voltage at low voltage side of step-up transformer	Not adjustable
PPC Rated Voltage [kV]	PPC control point rated voltage	User to configure accordingly
Rated Frequency [Hz]	System frequency	User to configure accordingly
Par_XL [ohm]	Reserved line compensation, please contact Envision if it is required	Not adjustable
Par_R [ohm]	Reserved line compensation, please contact Envision if it is required	Not adjustable
Sampling Frequency [Hz]	PPC sampling frequency	Not adjustable
Sampling Period [s]	PPC sampling period	Not adjustable
Filter Time Constant [s]	Reserved interface. Filtering dealt with internally	Not adjustable
Delay Time [s]	Data collection and communication delay	Not adjustable

Parameter	Description	Remark
<i>*Note: when the PPC model is in EMS dispatcher mode, it will bypass all the closed loop controllers and will directly dispatch P and Q commands to WTG or PCS; in this mode it therefore expects to receive P and Q reference from master PPC.</i> <i>**Note: as the PPC model has been implemented with fast initialization logic, to give enough time for initialization, setpoint change or disturbance should be applied not earlier than PPC Initialization Time +5s (and ensure that the BESS/WTG model is fully initialized) .</i>		

4.2 Reactive Power and Voltage Control

The PPC model provides 4 different reactive power control modes as shown in Table 2 below, user could select control mode via “Reactive Power Control Mode” under “QControlParam” sub-menu. When a certain control mode is selected, irrelevant parameter settings will be automatically greyed out to avoid confusion.

Table 2 Q/V control mode description

Control mode	Control function description
Q Control	The PPC targets to regulate control point reactive power to the setpoint value using a PI controller
Q/P Ratio Control	The PPC targets to regulate control point reactive power to the target value calculated the setpoint and control point active power, via the same control loop as for Q control mode
Power Factor Control	The PPC targets to regulate control point reactive power to the target value calculated from power factor setpoint value and control point active power, via the same control loop as for Q control mode
V Droop Control	The PPC targets to regulate control point reactive power to the target value calculated from voltage droop equation $Q_{target_kVAr} = (V_{ref_pu} - V_{meas_pu}) / K_{droop} * Base_kVA$, via a PI controller, when detected voltage deviation gets beyond deadband
V Loop Control	The PPC targets to regulate control point voltage to setpoint value via a PI controller, when detected voltage deviation gets beyond deadband

Table 3 below provides descriptions of the control parameters.

Table 3 Q/V control parameters description

Parameter Name	Setting Description	Applicable Control Mode
Reactive Power Control Mode	Control mode selection as described in Table 2	/
PCC Max Q Limit [kVAr]	Limitation on control point capacitive Q, positive value required	Q control, Q/P control, PF control, V droop control
PCC Min Q Limit [kVAr]	Limitation on control point inductive Q, negative value required	
QCmd RampRate Limitation [pu/s]	Ramp rate limit on reactive power command issued to WTG or PCS	All
Q Control Kp	Q control proportional gain	Q control,

Parameter Name	Setting Description	Applicable Control Mode
Q Control Ki	Q control integral gain	Q/P control, PF control
WT/BESS Qcapacity Factor	PPC to WTG/PCS Q command limitation in power factor@Pn format, Pn as kW rating of the WTG/PCS equipment	All
WT/BESS Terminal V Protect Factor [pu]	PPC freezes reactive power command sent to WTG/PCS if equipment voltage deviation based on 1pu exceeds the setting	All
V Droop Control Base [kVar]	“Base_kVA” as in V-Q formular $Q_{target_kVAr} = (V_{ref_pu} - V_{meas_pu}) / K_{droop} * Base_kVA$	V droop control
V Droop Control Slope	“K_droop” as in V-Q formular $Q_{target_MVar} = (V_{ref_pu} - V_{meas_pu}) * K_{droop} * Base_kVA$	V droop control
V Droop Control Positive Deadband [pu]	the positive dead-band of voltage for voltage droop control	V droop control V loop control
V Droop Control Negative Deadband [pu]	the negative dead-band of voltage for voltage droop control	V droop control V loop control
V Droop Control Kp	V droop control proportional gain	V droop control
V Droop Control Ki	V droop control integral gain	V droop control
V Loop Control Kp	V loop control proportional gain	V loop control
V Loop Control Ki	V loop control integral gain	V loop control

4.3 Active power Control

The PPC targets to regulate control point active power to the setpoint value using a PI controller; active power ramp rate can be applied if required. Parameters for active power control could be found in “PControlParam” of the PPC model menu as Table 4 shows below.

Table 4 P control parameters description

Parameter Name	Setting Description
P Limitation Coefficient [pu]	Plant active power higher limitation
Dispatch PCmd RampRate Limitation [pu/s]	AGC active power regulation command execution ramp rate limit
P Control Kp	P control proportional gain
P Control Ki	P control integral gain

4.4 Frequency Response

The PPC provides several different frequency response control modes and relevant parameters can be adjusted in “FrequencyResponseParam” tab of the PPC model menu.

In all different control modes when the control is triggered, the PPC regulates active power change of the plant as the same formular as below. The active power change is added to active power command and the same PI controller as for active power control is used.

$$\Delta P_{kW} = Base_kW * \Delta f_{Hz} / (50 * Droop)$$

Where:

ΔP represents the active power adjustment for frequency response and the unit is kW;

Δf represents the system frequency variation with deadband considered, and the unit is Hz;

Droop is the parameter frequency droop factor;

Base is the real value calculation base in kW.

Table 5 below describes how the frequency control mode should be configured in regards to different grid requirements.

Table 5 Frequency control mode configuration guide

Control requirement	Configuration description
Primary frequency response (PFR)	Put "Frequency Response Control Mode" as "FSM" and "FFR Enable" as "Disable"
Frequency sensitive mode (FSM)	
Limited frequency sensitive model - over/under (LFSM)	Put "Frequency Response Control Mode" as "LFSM-O/U"
Both FSM and LFSM on*	Put "Frequency Response Control Mode" as "FSM//LFSM"
Both FSM and LFSM on, and during LFSM the delta P calculated from FSM and LFSM are summed up	Put "Frequency Response Control Mode" as "FSM+LFSM"
(UK BESS) LFSM with Deload	Put "Deload Enable" as "Enable" while LFSM function is selected in "Frequency Response Control Mode". Deload function will then be automatically triggered with latching when BESS is in import mode under LFSM-U
(India) PFR and FFR	Put "Frequency Response Control Mode" as "FSM" to represent PFR function, and "FFR Enable" as "Enable"
*Note: When both FSM and LFSM on, LFSM stage active power change will be the summation of delta P calculated from FLSM and the FSM maximum delta P.	

More parameters for frequency control could be found in the menu when certain control mode is selected, as shown in Figure 3 below. Deadband, P-f droop and delta P limit for each control mode can be tuned as the parameter names indicate straightaway. A few other parameters are described in Table 6.

FrequencyResponseParam	
Frequency Response Control Mode	FSM
P-f factor calculation Base [kW]	55000
Minimum P for Trigger FreqRes [pu]	-1
FreqRes PCmd RampRate Limitation [pu/s]	1
FSM Frequency Response P-f Factor	0.05
FSM Positive Deadband [Hz]	0.5
FSM Negative Deadband [Hz]	-0.5
FSM deltaP UpLimit [pu]	2
FSM deltaP DnLimit [pu]	-2
LFSM-O Frequency Response P-f Factor	0.05
LFSM-O Positive Deadband [Hz]	0.4
LFSM-O deltaP DnLimit [pu]	-2
LFSM-U Frequency Response P-f Factor	0.1
LFSM-U Negative Deadband [Hz]	-0.5
LFSM-U deltaP UpLimit [pu]	2
FFR Enable	Disable
FFR Positive DeadBand [Hz]	0.3
FFR Negative DeadBand [Hz]	-0.3
FFR PCmd RampRate Limitation [pu/s]	1
Deload Enable	Enable
Deload Frequency Response P-f Factor	0.006
Deload Negative Deadband [Hz]	-0.5
Deload deltaP UpLimit [pu]	2
Deload PCmd RampRate Limitation [pu/s]	1
Latching Function Enable	Enable
FreqRes Reverse Limitation Enable(for PCS)	Enable

Figure 3 PPC frequency control parameters

Table 6 Frequency response parameters (in addition to deadband, droop and deltaP limit) description

Parameter Name	Setting Description
Frequency Response Control Mode	Control mode selection as described in Table 5
P-f factor calculation Base [kW]	“Base” as in Δf - ΔP formular “ $\Delta P_{kW} = \text{Base}_{kW} \cdot \Delta f_{Hz} / (50 \cdot \text{Droop})$ ”
Minimum P for Trigger FreqRes [pu]	Minimum P level that the plant can respond to frequency response and minimum P level that the plant can be curtailed to in over-frequency response
FreqRes PCmd RampRate Limitation [pu/s]	Frequency response P command execution ramp rate limit (separate ramp rate settings are provided for FFR and Deload mode)
FFR PCmd RampRate Limitation [pu/s]	FFR mode P ramp rate limit
Deload PCmd RampRate Limitation [pu/s]	Deload mode P ramp rate limit
Latching Function Enable	This switch works for all control modes except for Deload mode, which is hardcoded with latching function enabled. Enable = once frequency response is triggered, plant delta P does not decrease with delta f decreases, until frequency is back within deadband; Disable = plant active power dynamically changes with delta f when frequency is out of deadband range.
FreqRes Reserve Limitation Enable (for PCS)	Enable = BESS plant is allowed to change between export mode and import mode during frequency response; Disable = BESS plant is only allowed to operate in the same mode (export/import) during frequency response.

4.5 Inertia Response

When inertia response function is enabled, the plant will regulate change of active power based on rate of change of frequency (RoCoF) according to the following formula, and the active power change is added to active power command to help suppress the system RoCoF:

$$\Delta P_{kW} = \text{Base}_{kW} * (-Tj_s * df/dt_{Hz/s}) / 50$$

The relevant parameters can be adjusted under “InertiaResponseParam” tab and are described in Table 7 below.

Table 7 Inertia response parameters description

Parameter Name	Setting Description
Inertia Response Enable	Function switch
Inertia Time Constant(Tj) [s]	Equivalent inertia time constant
IR Positive Deadband [Hz/s]	Positive RoCoF deadband for inertia response
IR Negative Deadband [Hz/s]	Negative RoCoF deadband for inertia response
IR deltaP UpLimit [pu]	Increase P regulation amount limit
IR deltaP DnLimit [pu]	Decrease P regulation amount limit
IR PCmd RampRate Limitation [pu/s]	P ramp rate limit for inertia response

4.6 Voltage lock logic

The PPC is implemented with high voltage and low voltage lock logic.

PPC control point phase voltage is used in the detection, when either of the phase voltages exceeds high/low voltage lock on level for delay time, PPC locks the P and Q commands issued to equipment lumps, until all the phase voltages at control point is recovery back to within high/low voltage lock off level for delay time.

The relevant parameters can be adjusted under “PPCLockParam” tab as the parameter names indicate straightaway. Working as threshold hysteresis, “Low Voltage Lock Off Level” should be higher than “Low Voltage Lock On Level”, “High Voltage Lock Off Level” should be lower than “High Voltage Lock On Level”.

4.7 SVG Control

The PPC model provides SVG control function for up to 4 SVGs; however, please confirm with Envision Energy on product feasibility for SVG control before conducting simulation studies.

4.7.1 SVG dispatch logic

The PPC model provides three different modes on reactive power command dispatch when PPC controlling SVG. The dispatch mode can be selected via ‘Q Distribute Priority’ setting in the PPC menu, and dispatch logic in each mode is described as below:

- SVG first
 - SVG are dispatched in priority; if SVGs' capacity is not enough, reactive power of the wind turbines will then be regulated.
- WT/BESS first
 - WT/BESS are dispatched in priority; if WT/BESSs' capacity is not enough, reactive power of the SVG will then be regulated.
- Proportional (by installed Q capacity)
 - WT/BESS and SVG are dispatched equally in proportion to the WT/BES and SVG installed Q capacity.

With either of the modes, when there are more than one SVG, the dispatch among different SVGs will be per unit equal, based on each SVG's rating.

4.7.2 SVG control inputs and outputs

The PPC model gives 4 sets of SVG interfaces while each of the sets is for interface of one SVG and the first set of interface is labelled by default for reference. Each of the SVG input is a two-dimensional interface, with the first one as SVG enabled switch (0 as disabled, 1 as enabled) and the second one as the SVG rating in kVar (rating must be positive). The relevant output signal gives SVG reactive power command in kVar. Figure below gives two examples of controlling 2 SVGs and 3 SVGs respectively. The fourth set of the interfaces can be set in the same way if there are 4.

Example 1: PPC controlling two SVGs, SVG 1 rated at 70MVar and SVG 2 rated at 25MVar.

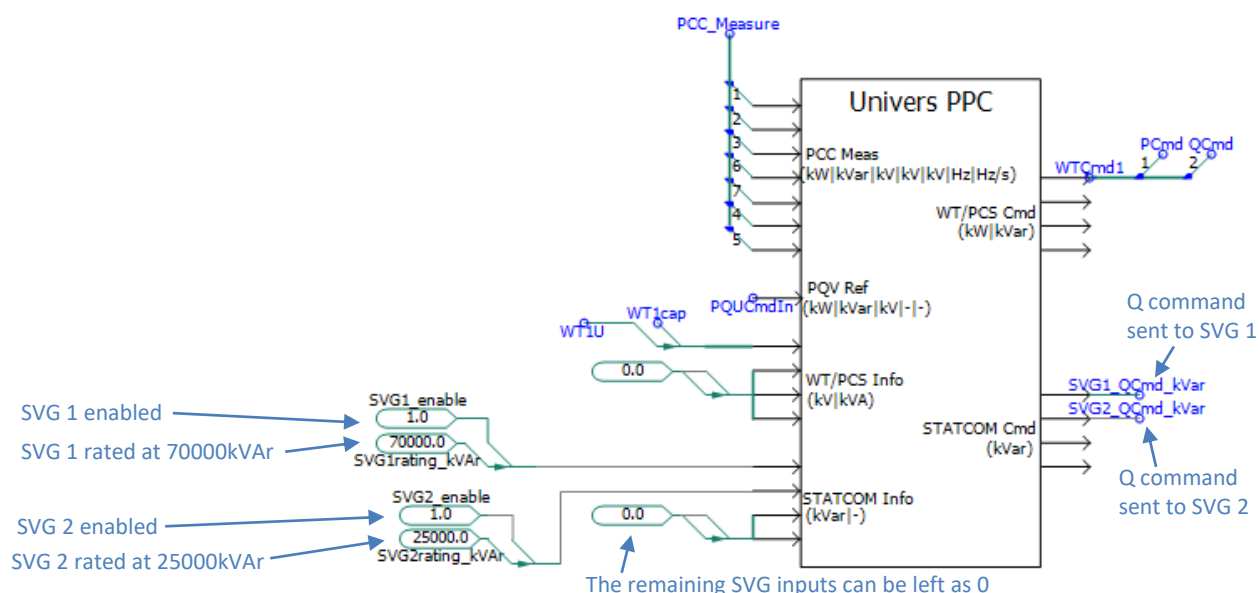


Figure 4 Interface setting example of controlling 2 SVGs

Example 2: PPC controlling three SVGs rated at 70MVar, 50MVar and 25kVar respectively.

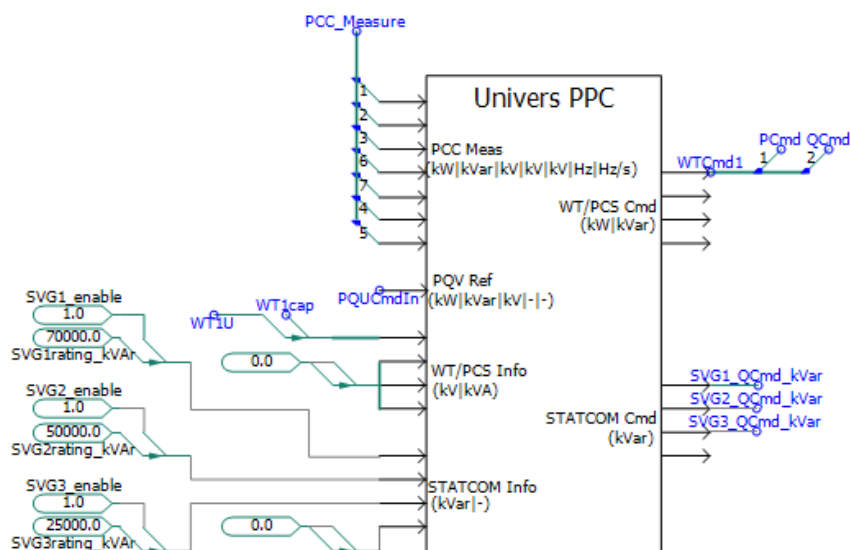


Figure 5 Interface setting example of controlling 3 SVGs